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Centre Number

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Student Number

2022 YEAR 12

Advanced Mathematics

Task 1 Validation Task

05/12/22

Q	Marks
1	/2
2	/2
3	/6
4	/2
5	/2
6	/5
7	/2
8	/6
9	/3
Total	/30

General

Instructions:

- Reading time – 5 minutes
- Working time – 35 minutes
- Write using blue or black pen
- NESA approved calculators may be used
- Show relevant mathematical reasoning and/or calculations
- No white-out may be used

Total Marks: 30

This question paper must not be removed from the examination room.

*This Validation task constitutes **60%** of the Assessment Task.*

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In Questions 1 – 9, your response should include relevant mathematical reasoning and/or calculations.

Question 1 (2 marks)

If $\log_{10} 2 = 0.3010$ and $\log_{10} 3 = 0.4771$, find:

(a) $\log_{10} 6$

1

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(b) $\log_{10} \frac{2}{9}$

1

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Question 2 (2 marks)

Solve the following using change of base. Write your answer to three significant figures.

2

$$5^x = 3$$

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Examination continues on next page

Question 3 (6 marks)

Solve for x if:

(a) $121^x = 11$

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(b) $5^{x+3} = 25$

1

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(c) $\log_2(x + 1) = 3$

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(d) $2\log_{10} x + 3 = \log_{10} x^5$

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Question 4 (2 marks)

Solve the following for x :

2

$$81^{x^2} = 3^{-4x - 1}.$$

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Question 5 (2 marks)

Phillip has placed \$2 000 000 in an interest-bearing account. He invested at a rate of 5% p.a. Using the compound interest formula shown below, determine the number of years (n) it will take for Phillip to at least double his money.

2

$$A = P(1 + r)^n$$

A (**Amount**) = *Principal plus interest*

P = *Principal*

r = *Interest rate as a percentage*

n = *time in years*

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Question 6 (5 marks)

A logarithmic scale is used to measure the sound level with units in decibels.

$l = 10 \log_{10} \frac{I}{I_0}$, where l is in decibels, I is the intensity of the sound in (watts/m²) and I_0 is the threshold of sound intensity detected by the human ear.

- (a) What is the decibel rating of sound at the threshold of hearing (i.e. $I = I_0$). **1**

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- (b) Find the sound level of a conversation, which is 1 million times more intense than the threshold of human hearing. **1**

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- (c) How much greater is the noise intensity of an iPod playing at a peak volume of 115dB (I_1) compared with a half volume level of 85 dB (I_2)? **3**

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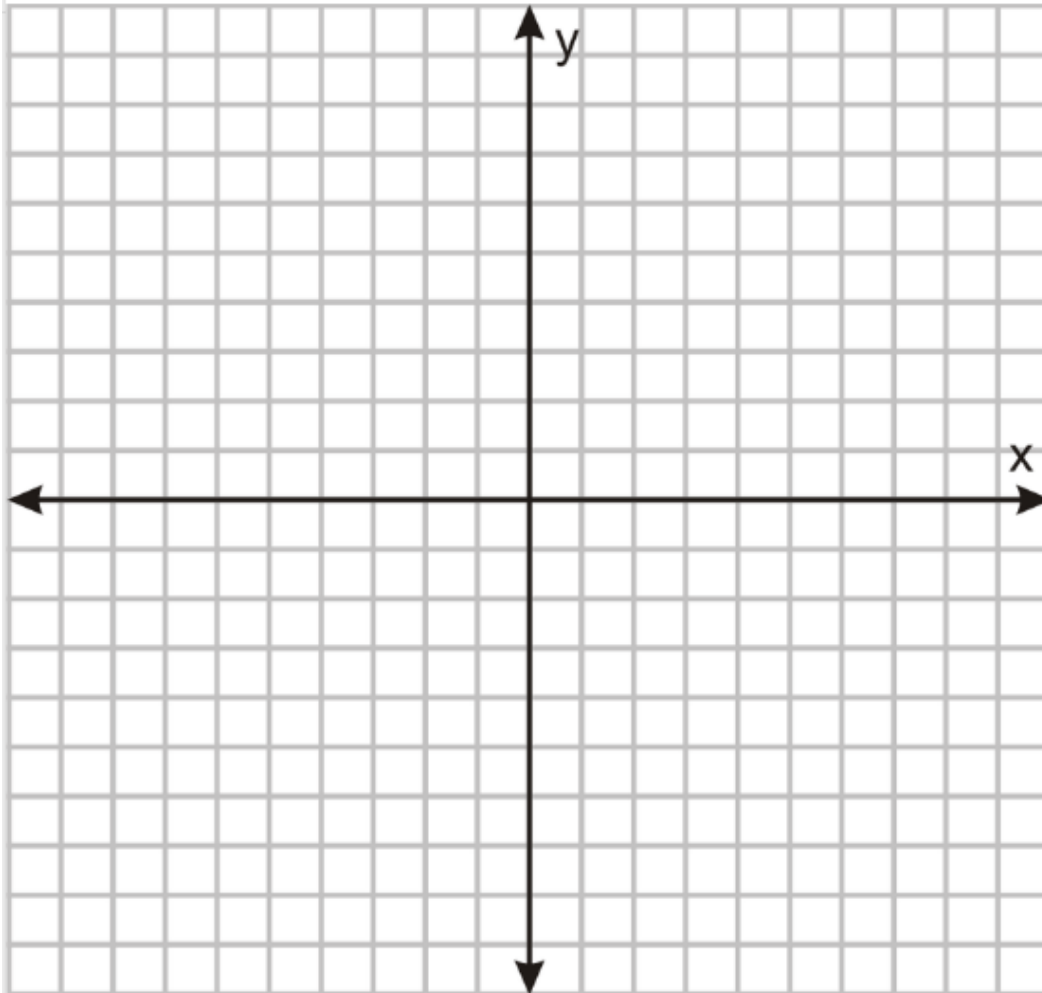
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Question 7 (2 marks)

Sketch the graph of the following exponential function in the box below. Label any intercepts and the asymptote. 2

$$y = 3e^{-x} + 1$$



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Question 8 (6 marks)

Differentiate the following, simplifying where necessary.

(a) $y = (x^2 + x + 1)e^{2x}$

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(b) $y = e^{x - \frac{1}{x}}$

2

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(c) $y = \frac{3 + e^x}{e^{4x}}$

2

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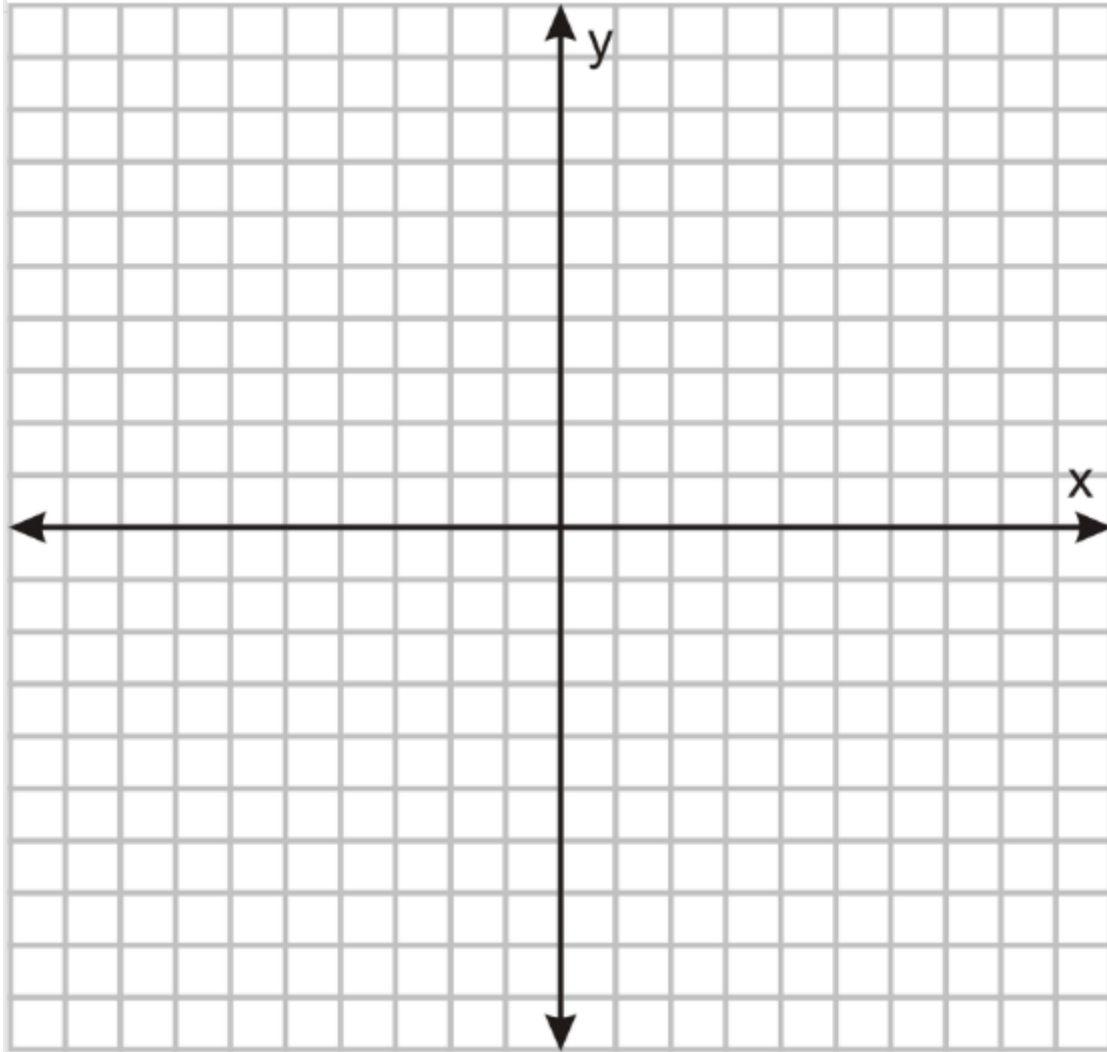
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Question 9 (3 marks)

Sketch the following graph, clearly indicating the vertical asymptote and labelling all critical points. **3**

$$-y = \log_e(x - 2) + 1$$



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END OF EXAMINATION

Examination continues on next page

Extra writing space

If you use this space, clearly indicate which question you are answering.

[illegible]

1.	Solutions and Marking Schemes	Markers Feedback
1(a)	Question 1 (2 marks) If $\log_{10} 2 = 0.3010$ and $\log_{10} 3 = 0.4771$, find: (a) $\log_{10} 6$ 1 $= \log_{10} 2 + \log_{10} 3$ $= 0.3010 + 0.4771$ $= 0.7781$ ① - mark for correct answer with working	
1(b)	(b) $\log_{10} \frac{2}{9}$ 1 $= \log_{10} 2 - \log_{10} 9$ $= 0.3010 - 2 \times 0.4771$ $= -0.6532$ ① - mark for correct answer with working	Some confusion with power rule.
2	Question 2 (2 marks) Solve the following using change of base. Write your answer to three significant figures. 2 $5^x = 3$ $x \ln 5 = \ln 3 \rightarrow$ This is not using change of base $x = \frac{\ln 3}{\ln 5} \rightarrow$ They must show this line for one mark. $x = 0.683$ ① mark for correct answer	Must show change of base in your working, as stated in the question.

3(a)	Solve for x if: (a) $121^x = 11$ 1 $11^{2x} = 11^1$ $2x = 1$ $x = \frac{1}{2}$ or $x = \frac{\log_e 11}{\log_e 121}$ $x = 0.5$ ① - mark for answer.	Some put $x = 2$. Double check your work.
3(b)	(b) $5^{x+3} = 25$ 1 $5^{x+3} = 5^2$ $x+3 = 2$ $x = -1$ ① - mark for correct answer.	
3(c)	(c) $\log_2(x+1) = 3$ 2 $x+1 = 2^3$ $x+1 = 8$ $x = 7$ - ① mark for w/o. - ① mark for correct answer (ECF applies.)	
3(d)	(d) $2\log_{10} x + 3 = \log_{10} x^5$ 2 $2\log_{10} x + 3 = 5\log_{10} x$ $3 = 3\log_{10} x$ $\log_{10} x = 1$ $x = 10$ - ① mark for w/o - ① mark for correct answer.	Some added brackets which changed the question to $2(x+3) = x^5$

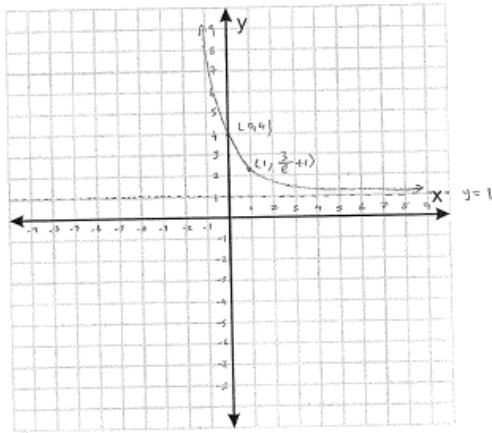
4	Solve the following for x: 2 $81x^2 = 3^{-4x-1}$ $3^{4x^2} = 3^{-4x-1}$ $4x^2 = -4x - 1$ $4x^2 + 4x + 1 = 0 \rightarrow \textcircled{1} - \text{mark}$ $(2x+1)^2 = 0$ $\therefore x = -\frac{1}{2} \quad \textcircled{1} - \text{mark for correct answer}$	Some were stuck on factorising the non-monic quadratic.
5	Phillip has placed \$2 000 000 in an interest-bearing account. He invested at a rate of 5% p.a. 2 Using the compound interest formula shown below, determine the number of years (n) it will take for Phillip to at least double his money. $A = P(1+r)^n$ A (Amount) = Principal plus interest P = Principal r = Interest rate as a percentage n = time in years $4\,000\,000 = 2\,000\,000 (1+5\%)^n$ $2 = 1.05^n$ $\log 2 = n \log 1.05$ $n = \frac{\log 2}{\log 1.05}$ $n = 14.20669985 \dots \rightarrow \textcircled{1} \text{ mark for m.f.o.}$ $n = 15 \text{ years} \rightarrow \textcircled{1} \text{ mark for rounding to the next year.}$	Poorly done. - Many assumed that $5\% = 5$ - Many did not know how to set up the initial equation - Many did not know to round up to an integer, as interest is only compounded at the end of a year.

6(a)	A logarithmic scale is used to measure the sound level with units in decibels. $l = 10 \log_{10} \frac{I}{I_0}$, where l is in decibels, I is the intensity of the sound in (watts/m²) and I_0 is the threshold of sound intensity detected by the human ear. (a) What is the decibel rating of sound at the threshold of hearing (i.e. $I = I_0$). 1 at $I = I_0$ $10 \log_{10} \frac{I_0}{I_0}$ $= 10 \log_{10} 1$ $= 10 \times 0$ $= 0$ — (1) mark w/o ...	
6(b)	(b) Find the sound level of a conversation, which is 1 million times more intense than the threshold of human hearing. 1 $I = 10^6 \cdot I_0$ $10 \log_{10} \left(\frac{10^6 \cdot I_0}{I_0} \right) = 2$ $= 60 \text{ dB}$ (1) mark for substituting $I = I_0$ and an answer of 60 dB	
6(c)	(c) How much greater is the noise intensity of an iPod playing at a peak volume of 115dB (I_1) compared with a half volume level of 85 dB (I_2)? 3 $L_1 = 115 = 10 \log_{10} \left(\frac{I_1}{I_0} \right)$ $L_2 = 85 = 10 \log_{10} \left(\frac{I_2}{I_0} \right)$ $11.5 = \log_{10} \left(\frac{I_1}{I_0} \right)$ $8.5 = \log_{10} \left(\frac{I_2}{I_0} \right)$ $\frac{10}{10} 11.5 = \frac{I_1}{I_0}$ $\frac{10}{10} 8.5 = \frac{I_2}{I_0}$ $I_0 = \frac{I_1}{10^{11.5}}$ (1) mark $I_0 = \frac{I_2}{10^{8.5}}$ (1) mark $\frac{I_2}{I_1} = \frac{10^{8.5}}{10^{11.5}} = 10^{-3}$ (1) mark $\frac{I_2}{I_1} = 10^{-3}$ answer. How much greater is the noise intensity of an iPod playing at a peak volume of 115dB (I_1) compared with a half volume level of 85 dB (I_2)? L	

7

Sketch the graph of the following exponential function in the box below. Label any intercepts and the asymptote. 2

$$y = 3e^{-x} + 1$$



There are no x intercepts.

① - mark for labelling the y intercept

① - mark for a neat sketch

8(a)

Differentiate the following, simplifying where necessary.

(a) $y = (x^2 + x + 1)e^{2x}$

$$\frac{dy}{dx} = u'v + uv'$$

$$= e^{2x}(2x+1) + (x^2+x+1) \cdot 2e^{2x}$$

$$= (2x^2 + 2x + 2)e^{2x} + (2x^2 + x)e^{2x}$$

$$= e^{2x}(2x^2 + 4x + 3) \quad \text{--- ① mark for simplifying correctly.}$$

u/v.

$$u = x^2 + x + 1$$

$$u' = 2x + 1$$

$$v = e^{2x}$$

$$v' = 2e^{2x}$$

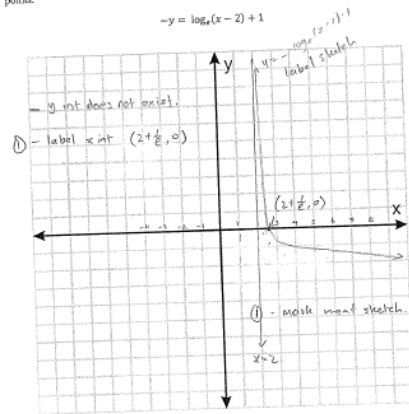
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① - mark for u/v.

8(b)	(b) $y = e^{x-\frac{1}{x}}$ let $u = x - \frac{1}{x}$ $\frac{dy}{du} = e^{u-\frac{1}{x}} \cdot (1 + \frac{1}{u^2})$ $\frac{du}{dx} = 1 + \frac{1}{x^2}$ $\frac{dy}{dx} = (1 + \frac{1}{x^2}) e^{x-\frac{1}{x}}$ $y = e^u$ $\frac{dy}{dx} = e^u \cdot \frac{du}{dx} = e^{x-\frac{1}{x}} \cdot (1 + \frac{1}{x^2})$ ① ans 1	2
8(c)	(c) $y = \frac{3+e^x}{e^{4x}} \cdot \frac{u}{v}$ $u = 3+e^x$ $u' = e^x$ $v = e^{4x}$ $v' = 4e^{4x}$ $\frac{dy}{dx} = \frac{vu' - uv'}{v^2}$ $= \frac{e^{4x} \cdot e^x - (3+e^x) \cdot (4e^{4x})}{(e^{4x})^2}$ $= \frac{e^{5x} - 12e^{4x} - 4e^{5x}}{e^{8x}}$ $= \frac{-11e^{5x} - 4e^{5x}}{e^{8x}}$ — ① correct numerator $= \frac{-15e^{5x}}{e^{8x}}$ — ① correct denominator or $y = 3e^{-4x} + e^{-3x}$ $y' = -12e^{-4x} - 3e^{-3x}$ $= -\frac{12}{e^{4x}} - \frac{3}{e^{3x}} - 8 -$	2

9

Sketch the following graph, clearly indicating the vertical asymptote and labelling all critical points.



$$y = -\log_2(x-2) - 1$$